

WTW 158 Worksheet 1 on inequalities and absolute value

A TEXTBOOK PROBLEMS

p A9: 17, 19, 27, 29, 35

p A9: 7, 12, 45, 49, 51, 55

B. MORE PROBLEMS - Solve: (Answer in interval notation)

1. $-3(r-4) \geq 0$

2. $m-2 < -8$ or $-\frac{m}{8} < -1$

3. $x+8 \geq 9$ and $2-x > -1$

4. $\frac{3-x}{x+1} \leq 0$

5. $\frac{|4x|}{x-3} < 0$

6. $\frac{1-x}{(x+2)(2-x)} \geq 0$

7. $\frac{x-1}{x} < 4$

8. $x|x| < 0$

9. p 39 Studyguide Technique mastering A - Absolute value

Do the following: 2 a, b, c, i, k, m

10. Write out in the form $f(x) = \begin{cases} \dots & \text{if } / \text{ as } x \in \dots \\ \dots & \text{if } / \text{ as } x \in \dots \end{cases}$ and sketch the function

(i) $f(x) = |4 - x^2|$

(ii) $f(x) = x|x|$

C. SELECTED TEST QUESTIONS

Question 1

Solve for x : $1 - \frac{1}{1-x} > 0$

Question 2

The solution set of $|x-1| > -1$ is

[2a] $\emptyset$ (empty set)	[2b] $\mathbb{R}$	[2c] $(0, \infty)$	[2d] $(-\infty, 2)$
[2e] $(-\infty, 0) \cup (2, \infty)$	[2f] $(0, 2)$	[2g] None of these	

Question 3

i Write down the definition of the absolute value of a real number x .

ii Write $\sqrt{|x-1|+4}$ in the form $\begin{cases} \dots & \text{if } x \geq \dots \\ \dots & \text{if } x < \dots \end{cases}$

iii Sketch the graph of the function $f(x) = \sqrt{|x-1|+4}$.

D. YOUTUBE VIDEOS - for extra exposure

Search:

1. Solving rational inequalities Mario's Math Tutoring

2. Solving Absolute Value Inequalities Mario's Math Tutoring

SELECTED ANSWERS

A. 17. $x \in (3, \infty)$, 19. $x \in (2, 6)$, 27. $x \in [-1, \frac{1}{2}]$, 35. $x \in (-1, 0) \cup (1, \infty)$

7. $2-x$, 45. $x \in \{2, -\frac{4}{3}\}$, 49. $x \in (3, 5)$, 51. $x \in \mathbb{R} \setminus (-7, -3)$

B. 1. $r \in (-\infty, 4]$, 2. $m \in \mathbb{R} \setminus [-6, 8]$, 3. $x \in [1, 3)$, 4. $x \in \mathbb{R} \setminus [-1, 3)$, 5. $x \in (-\infty, 3) \setminus \{0\}$,

6. $x \in (-2, 1] \cup (2, \infty)$, 7. $x \in \mathbb{R} \setminus [-\frac{1}{3}, 0]$, 8. $x \in (-\infty, 0)$, 9. 2a. $x = 2$, 2b. $x \neq 2$,

2c. $x \in \emptyset$, 2i. $x \in \{-1, 2\}$, 2k. $x \in \emptyset$, 2m. $x \in \{-3, 3\}$